Product Requirement Document (PRD) — Stitch AI Data Suite

Document Version: 1.0.0
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Status: Approved

1. Executive Summary

Stitch is an enterprise-grade, privacy-first, session-isolated AI Data Suite designed to democratize advanced data intelligence and automated machine learning. By utilizing local session memory and localized metadata context extraction, Stitch provides business stakeholders and data analysts with powerful tools to clean, analyze, visualize, and model data without exposing raw or sensitive data to external servers. This document establishes the comprehensive product, functional, and non-functional requirements for the Stitch platform.

2. Product Vision & Value Proposition

Traditional business intelligence and automated machine learning tools require extensive data pipeline overhead or pose severe data governance and compliance risks when uploading data to cloud services. Stitch delivers a zero-setup, zero-data-leakage client environment that combines deterministic programmatic scripts with large language model narration. Key value propositions include:

- **Absolute Privacy:** Raw datasets are kept isolated within the user's volatile session state. Only metadata and top aggregate summaries are ever transmitted to external large language models.
- **Unified Analytics Lifecycle:** A single platform that guides the user completely from raw data ingestion to auto-cleaning, conversational querying, model comparison, and executive PDF reporting.
- **Accessible Advanced Analytics:** Natural language processing maps directly to Python code execution environments, allowing non-technical users to query data frameworks effortlessly.

3. Functional Requirements Matrix

The functional lifecycle of Stitch is divided into seven core functional modules. The table below highlights the module specifications and priorities:

Module Name	Core Functional Requirements	User Interface Element	Priority
1. Data Ingestion	Support CSV, XLSX, XLS, and JSON file parsing up to 200MB. Ensure local parsing validation.	Drag-and-Drop dashzone with size validation chip.	Critical
2. Data Cleaning	Automated dropping of columns exceeding 60% missing data threshold. IQR outlier bounding box evaluation.	Interactive execution checkboxes with preview toggle.	High
3. Analytics (EDA)	Generate automated metrics (Quality Score, Missing rows percentage, duplication flags) and high-performance Plotly visual elements.	Dark-mode interactive layout grid with selectbox adjustments.	High
4. AI Insights	Trigger generative prompts to produce automated summaries, descriptive guides, and automated local PII exposure audits.	Tabbed dashboard views with custom markdown presentation elements.	Medium
5. Chatbot Core	Parse queries natively via pattern matching scripts. Pass complex descriptive queries to active generative	Persistent dialogue chat interface with recommended prefill chips.	High

Module Name	Core Functional Requirements	User Interface Element	Priority
	AI models.		
6. AutoML Workspace	Auto-infer classification or regression parameters. Train and cross-validate across multiple algorithmic architectures (RF, Ridge, XGBoost).	Configuration expanders with metric progress counters.	Medium
7. Reports Layer	Compile structural metadata, visual configurations, and metric snapshots into exportable PDF or Excel files.	Actionable export download panels.	High

4. User Interface & Core Design System Guidelines

The platform requires a premium, dark-mode design token system that increases emphasis on data visualization clarity. The design system enforces:

- **Background Layout:** Ultra-deep base tones paired with neomorphic surfaces for distinct sections.
- **Typography:** Professional, highly legible clean fonts with mono-spaced formatting constraints applied selectively to technical metrics.
- **Card Design:** Glassmorphism elements utilizing light transparency rules and slight inner borders to highlight core key metrics.

5. Non-Functional Requirements

To meet the operational needs of enterprise environments, the platform satisfies the following performance requirements:

- **Security compliance:** Zero persistent cloud tracking of transactional user content. Absolute session state destruction upon tracking closure or disconnect events.

- **Performance latency bounds:** Local processing scripts must execute within 500 milliseconds for standard data manipulations on baseline files. AI interactions must stream gracefully using active placeholder spinners.
- **Robust Fallback Modes:** If an active Large Language Model connection falls offline or hits API quotas, pages must cleanly gracefully degrade to native local processing components.